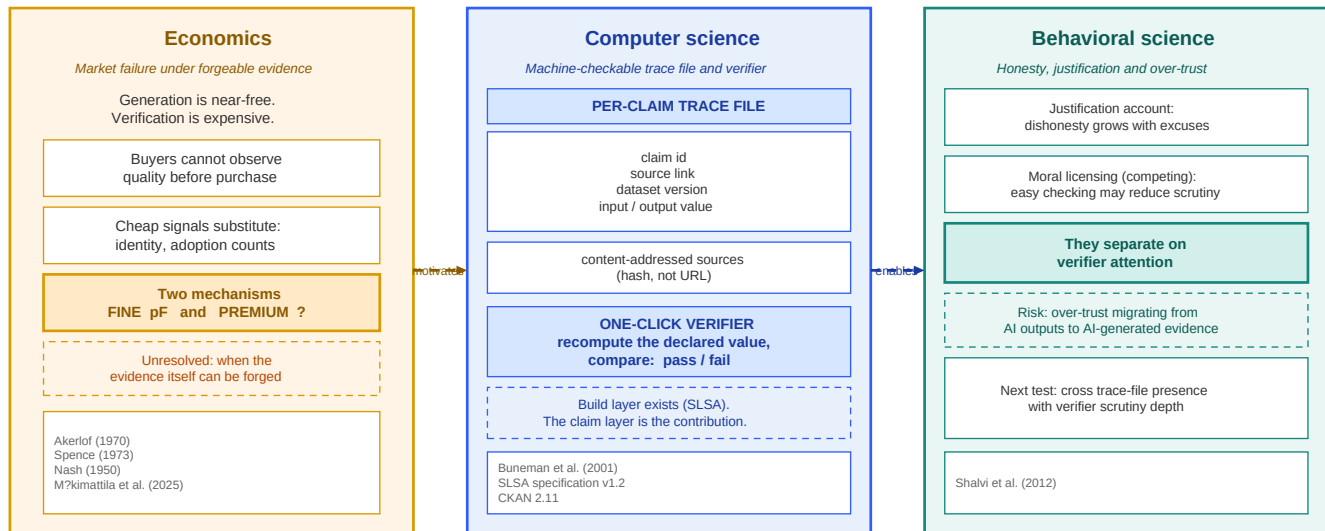


Honesty Through Two-Way Verification: A Credible Disclosure Infrastructure for the Era of Free Generation

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line, its mechanism, and the works cited for it. Solid boxes are established relations; dashed boxes and dashed arrows are the proposed synthesis. All values are simulated under the payoff law in the Colab notebook.

Figure 1: Prospective research design. Three disciplines, one strategic problem, with the works cited for each shown at the foot of its column. Free generation makes fabrication of the documents that carry information nearly costless, so the economic question shifts from what should be disclosed to which rules make honest disclosure stable [1, 6, 7, 10]. A machine-checkable per-claim trace file makes a declaration verifiable at near-zero cost, extending the build-layer provenance that already exists [2, 9]. Because effort-free checking may license inattention, the behavioral question is whether traceability shrinks the space for self-serving justification or invites over-trust [8]. Solid boxes are established relations; dashed boxes and dashed arrows are the proposed synthesis. Simulated outcomes, not observations of cognition or markets.

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Course session: Autumn 2026 Session 1. **Workshop:** Workshop 6.

Instructor: Luyao Zhang. **Assignment:** PS1 research proposal.

1 Three connected research questions

Generating a deliverable, and the documentation meant to vouch for it, is now nearly free, while independent verification remains costly. When anyone can fabricate the evidence, the question is not what to disclose but which rules make truthful disclosure stable. Stakeholders are plugin-marketplace developers, the platforms setting the rules, and the users and journalists who must act on a claim.

Three foundations frame the problem. Akerlof [1] shows unverifiable quality drives good sellers out of a market. Shalvi et al. [8] show dishonesty grows with available justifications. Buneman et

al. [2] define provenance as structured lineage, but the producer writes the record, so the trace itself can be forged.

Q1—Economics: Which mechanisms—a fine pF on detected fakes, or a market premium Δ for verifiable provenance—make the honest strategy a Nash equilibrium when trace files are free to generate and to fake?

Q2—Computation: What trace format and verifier make a per-claim declaration checkable at near-zero cost, without trusting the producer’s own record?

Q3—Behavior: Does traceability reduce dishonesty by shrinking the justification space, or does effortless checking produce moral licensing that increases it?

The disciplines are jointly necessary: a rule requiring trace files is empty unless a fake can be cheaply detected, and a rule that is enforceable may still fail if checking licenses inattention. Figure 1 connects the three.

2 Answer to Q1: incentives and intellectual lineage

Hypothesis. Two mechanisms can make honesty optimal, and either suffices. A penalty pF falls only on a detected fake; a premium Δ is earned only by a verifiable trace file. Honesty is the unique best response when $\Delta + pF$ exceeds the fake’s cost advantage.

Model. Developers choose an action $a_i \in \{\text{honest, none, fake}\}$ and verifier enrolment $v_i \in \{0, 1\}$; a platform commits to (m, p, F) for mandatory attachment, spot-check rate and fine, and the market sets a premium Δ for verifiable provenance. The platform sets rules, developers act, detected fakes pay F , and buyers pay Δ only for a trace they can check. Payoff is market benefit plus premium if honest, minus cost, plus stability if enrolled, minus expected fine if faking. A fake earns the same base market benefit as an honest deliverable, because buyers cannot observe truth before purchase, while saving cost. Information is incomplete. The solution concept is Nash equilibrium in pure strategies [7], with (m, p, F) exogenous; since the platform chooses the rules that developers best-respond to, this is a mechanism-design problem. I distinguish inherited quantities (Akerlof’s benchmark), simulated quantities (this payoff law) and hypothetical ones (real platforms).

Table 1: Strategy matrix with no rules ($p = F = 0$). Left: no premium, so faking wins. Right: a premium for verifiable provenance, so honesty wins without any enforcement.

		$\Delta = 0$	$\Delta = 30$
A	honest trace	135	165*
B	no trace	120	120
C	faked trace	160*	160

Without a premium, faking wins by 25—exactly the cost it avoids—and only a fine can deter it: $p^* = 0.5$ at $F = 50$. With a premium, honesty wins by being worth more and needs no enforcement. The mechanisms substitute: $\Delta = 15$ halves the required spot-check rate to 0.2, and $\Delta = 25$ removes the need to fine at all.

What the benchmark leaves unresolved. Akerlof shows unverifiable quality drives good sellers out of a market, but fixes buyers’ beliefs rather than asking how a costly signal forms them. Spence [10] supplies that logic: a premium exists only if the signal separates types. Neither says which rule restores honesty when the certification instrument is itself freely forgeable.

3 Answer to Q2: method and evaluation

Method. A browser-run 3×2 game on a Hugging Face Space, client-side HTML, CSS and JavaScript; the payoff law is documented in Appendix A. The algorithm enumerates the six cells and finds the best response.

Correction to the baseline. The first deployed version summed a cost, a benefit and a stability index. The notebook exposed the error: cost and benefit have opposite signs, so summing them scored honesty as better on both. The sweep returned $p^* = 0$, meaning faking was never attractive and the question could not be posed. The corrected law enters cost negatively and gives the fake the same market benefit as an honest deliverable. See Appendix B.

Baseline, modification, metric, check. Baseline is the status-quo cell (net 100); the modification is mandatory attachment plus p and F ; the metric is the threshold p^* at which honest-plus-verified is uniquely best. Five checks pass: with no rules the fake wins; with $p = 1, F = 100$ honesty is unique; the status quo is unchanged; enrolment is weakly dominant; there are six cells.

Result. At $F = 50$, $p^* = 0.5000$; the required p falls as F rises ($F = 100 \Rightarrow 0.25$; $F = 400 \Rightarrow 0.0625$), and below $F = 25$ no $p \leq 1$ suffices. All results are *simulated*.

Table 2: From laureate foundations to a new interdisciplinary contribution.

Horizon	Milestone and evidence
Foundation	Akerlof (1970): unverifiable quality collapses markets. Spence (1973): a credible signal needs single crossing. Mäkimattila et al. (2025): choice of certification is itself the signal.
PS1	A 3×2 game with a verified payoff matrix; two mechanisms located: a fine (pF) and a premium (Δ), with $\Delta + pF > 25$ required.
Next study	Endogenise Δ in buyer beliefs: test whether trace publication separates types, or pools and collapses. A test separating justification from licensing.
Longer term	A verifier a non-specialist can use, reporting which claims were checked, at what accuracy and what cost.
2056	Verifiable honesty that is competitively rewarded rather than merely enforced, and readable by any user.

4 Answer to Q3: behavior and competing explanations

Competing explanations. Justification: dishonesty grows with available justifications [8], so traceability should shrink that space. Moral licensing: a visible attestation lets producer and verifier treat the check as discharging responsibility, so the producer shades a claim the verifier no longer examines. They predict opposite effects from the same manipulation.

Discriminating evidence. They separate on verifier attention. Justification predicts less faking with scrutiny unchanged; licensing predicts reduced scrutiny and a fake rate falling less than the attachment rate rises. Measuring both producer choice and check depth discriminates; measuring the fake rate alone cannot.

Consequence. If licensing dominates, verification must not be a binary pass: it should report which claims were recomputed, and pF must be paired with something keeping scrutiny positive, since it deters only if p is real scrutiny.

Limits. A simulated table cannot establish human behavior. The next test is a short experiment crossing trace-file presence with check depth.

5 Advanced development and future directions

Foundation. Akerlof’s quality-uncertainty analysis [1], recognized by the 2001 Nobel Memorial Prize in Economic Sciences [11], establishes that unverifiable quality destroys markets and that institutions making quality observable restore trade. The enduring question is trust under asymmetric information. Spence [10] supplies the complementary logic: a costly credible signal separates quality, but only if the receiver can read it.

The boundary. Build-layer provenance [9] already exists. Unresolved is the claim layer: per-claim sources and input–output declarations, recomputed at near-zero cost, when the producer writes the record and can therefore forge it.

Where the premium must come from. Treating Δ as a constant merely relocates the assumption. A premium exists only if publishing a trace is informative about type, so that buyers update upward—the single-crossing condition that Spence [10] requires of any separating signal. If fakers can publish traces as cheaply as honest producers, the signal pools, Δ collapses to zero, and enforcement is again the only mechanism. Making Δ endogenous to buyer beliefs is therefore the next study, and the certification–design result of Mäkimattila et al. [6] is the natural instrument: there, what a sender reveals is not only the test outcome but the choice of which test to buy, and selection across tests is itself the signal. Its warning applies directly here—in that model the receiver obtains zero surplus, so a premium does not flow to honest producers automatically, and competition or rule design must be what directs it.

The verifier is built for machines. It recomputes a declared number and returns pass or fail, which a non-specialist cannot interpret. Letting an ordinary user judge a firm’s claim, or its data-handling practice, is a problem of cognitive rather than computational cost, where the objective is accurate enough and fast enough rather than perfect.

Expected 2056 Nobel Prize and Turing Award contribution statement. By 2056, I aim to contribute a governed trust infrastructure for an era of free generation, in which verifiable provenance is competitively rewarded rather than merely enforced, and in which any ordinary user can check a firm’s claims and data-handling practices without technical training. Building on Akerlof’s quality-uncertainty foundation, Spence’s signalling logic and machine-checkable provenance, I will integrate economics, computer science and behavioral science so that near-zero-cost trace files, a verifier designed for non-specialists, and platform rules that make honest publication an equilibrium together let anyone inspect a claim in under a minute. Progress will be shown by an endogenous premium that separates honest from faking producers, a verifier whose accuracy and cost are documented, a discriminating behavioral test, and a rule design adopted on a live platform.

Open Science Statement

GitHub: <https://github.com/peterhe0616-sketch/PS1-Zhengjun>

Google Colab: [Insert verified Colab URL]

Released: a browser-run simulator and a Colab notebook that re-derives the payoff table and runs the threshold sweep. The simulator is static client-side HTML/CSS/JavaScript with no backend, key or paid service; the notebook needs only Python. Inputs are synthetic: one payoff law read by both. No data are collected. Shortest run: open the notebook and choose *Run all*. Versions are in Appendix A.

Statement of Contribution to the UN’s SDGs

Hugging Face: <https://huggingface.co/spaces/Peterhe123/Trust>
Through an open-access game, this project supports **SDG 4—Quality Education** [12] by helping students and the public learn why honest disclosure requires cheap verification, by comparing the six rule-and-action combinations. No account or installation is needed. A feasible evaluation is a short pre/post question predicting which rule is stable. The resource released is the simulator; learning gains remain to be evaluated.

Submission milestones

First draft: September 13, 2026, 11:00 P.M. Beijing time (UTC+8).
Class review: September 14. Proposed: rebuttal September 16 and
revised submission September 20, both at 11:00 P.M.

Author Notes

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I thank Prof. Luyao Zhang, Program Chair, and Prof. Ken Rogerson, Program Discussant, of the *2056 Nobel and Turing Laureate Program Launch Workshop*, held at Duke Kunshan University on September 7, 2026. I also thank Longfei Jing and Yongzhuo Wu, my teammates in Workshop 6, and the rest of the COMSCI/ECON 206 Autumn 2026 Session 1 class. [Identify a specific contribution you actually used.] I am responsible for the proposal and remaining errors.

Individual contribution

I formed the core idea that in a market where AI makes generation free and verification expensive, every deliverable should carry a near-free, automatically generated trace file. I posed the research question, and developed the one-click verifier, the lemons-market framing and the rule triple (m, p, F) . I verified the relevant literature and compared my idea with existing work, and verified the game by playing it and checking its payoff logic, rules and report. On 9 September I found and corrected a sign error in the payoff law (Appendix B). The photographs, the game numbers and the final wording are mine.

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A Technical details and reproducibility

Symbol	Definition
a_i	Developer i 's action: honest trace, no trace
v_i	Whether developer i enrolls in the platform
m	Whether trace-file attachment is mandatory
p	Spot-check rate: probability a submitted trace is checked
Notation. F	Fine paid by a fake that is detected.
pF	Expected fine on a faked trace file.
Δ	Premium buyers pay only for a verifiable trace
p^*	Smallest p making honest the unique best response
b_a, c_a	Market benefit and cost of action a .
s	Stability gained by enrolling in the verifier

Artifact. Trace-or-Fake Market Game, a 3×2 provenance-rule simulator. Space: <https://huggingface.co/spaces/Peterhe123/Trust>. Files: `index.html`, `README.md`, `verify_game.mjs`. Author: Zhengjun He. v1.0.0, 6 September 2026.

What it does. A developer publishing a tool answers two questions. Q1, trace file: provide one (cost +5%, benefit +20%), none (no change), or fake one (cost −20%, benefit −20%). Q2, verifier: enrol (cost +5%, stability +20%) or not. All indices start at 100; the page prints a report after three runs. Nothing is stored and no network request is made.

Payoff law and checks. One block is read by both the game and the notebook, so they cannot disagree. Five checks pass: with no rules the fake wins; with $p = 1, F = 100$ honesty is unique; the status quo nets 100; enrolment is weakly dominant; there are six cells. The law carries two mechanisms: a fine pF charged only to a detected fake, and a premium Δ paid only for a verifiable trace file. Enrolment in the verifier costs 5 and adds 20 stability, so it is weakly dominant and the effective choice is among the three actions.

Correction, 9 September 2026. The first version summed a cost, a benefit and a stability index. Cost and benefit have opposite signs, so summing them let honesty win on both (105 vs 80 cost, 120 vs 80 benefit) before any rule applied, and the sweep returned $p^* = 0$: faking was never attractive, so the question could not be posed. The corrected law enters cost negatively and gives a fake the same market benefit as an honest deliverable. It now yields a real threshold: the fake wins with no rules (160 vs 135), and at $F = 50$ honesty requires $pF > 25$, so $p^* = 0.5000$; the required p falls as F rises ($F = 100 \Rightarrow 0.25$; $F = 400 \Rightarrow 0.0625$), and below $F = 25$ no $p \leq 1$ suffices.

Second correction, same date: the missing premium. With the law above, honest and faked traces were paid identically, so honesty was a tax rather than an asset and was never chosen for being worth more. A market premium Δ paid only for a verifiable trace file repairs this. The two mechanisms substitute exactly: honesty requires $\Delta + pF > 25$. At $\Delta = 0$ the fine must do all the work ($p^* = 0.5$ at $F = 50$); at $\Delta = 15$ the required rate halves to 0.2; at $\Delta = 25$ no enforcement is needed at all and honesty wins with $p = F = 0$. Values are from the verified notebook run; the deployed Space should be updated to this law.

Fresh-run record. Python 3.11, standard library only, no seed needed since both sweeps are deterministic. The notebook ran top to bottom with no error; abridged output:

```

A_honest 135   B_none 120   C_fake 160   (no
rules,  $\Delta=0$ )
Best response with no rules: C_fake
PASS no_rules_fake_wins, strong_rule_honesty_unique,
      status_quo_unchanged, enrolment_weakly_dominant,
six_cells
Threshold  $p^* = 0.5000$  ( $F = 50$ )   fake advantage
= +25.0
F=25: none   F=50: 0.5000   F=100: 0.2500   F=400:
0.0625
--- premium sweep,  $p = F = 0$  ---
delta=0: C_fake   delta=25: tie   delta=26: A_honest
delta=30: A_honest
--- combined threshold:  $\text{delta} + pF > 25$  ---
delta=0 ->  $pF=25.0$  ( $p^*=0.50$  at  $F=50$ )
delta=15 ->  $pF=10.0$  ( $p^*=0.20$  at  $F=50$ )
delta=25 ->  $pF= 0.0$  ( $p^*=0.00$ : no enforcement
needed)

```

Boundary. Static client-side HTML/CSS/JavaScript only: no runtime, backend, key or paid service. Hosting and the viewer’s browser still consume energy, and only the narrower claim is made [5]. Semantic fieldsets, table captions, aria-live regions and keyboard operability are present; a full screen-reader pass is unverified.

A.1 AI-use disclosure

Record. DeepSeek, model deepseek-v4-flash, 6 September 2026, for literature comparison and for debugging the game’s JavaScript. Accepted: that build-layer attestation already exists, so the contribution belongs at the claim layer. Rejected: framings claiming the artifact establishes human behavior. I re-read every source, re-derived the table, played the game and ran the checker. Initial reasoning and handwritten reflection are Human-Only.

B Cumulative Intellectual Development

This proposal develops my Intellectual Statement II through the *2056 Nobel and Turing Laureate Program Launch Workshop*, September 7, 2026, Duke Kunshan University, IB 2050. Program Chair: Prof. Luyao Zhang. Program Discussant: Prof. Ken Rogerson. My session: Workshop 6. My role: [assigned role]. My teammates: Longfei Jing and Yongzhuo Wu.

First change: the object of the question. My earlier claim was that I would build mechanisms letting agents and their designers cooperate under near-total unobservability. The September 4 field evidence forced something narrower. At the Tencent Shanghai Office, plugin quality is hard to observe before adoption, so buyers rely on cheap signals. At the Shanghai Science and Technology Museum, the deep-residual-learning panel asks a general audience to trust institutional authority rather than checkable evidence. The marketplace is a market without institutional trust; the museum is institutional trust without a market. Workshop feedback sharpened the boundary: build-layer provenance already exists at near-zero cost, so the contribution must sit at the claim layer. That moved the proposal from a disclosure design to a rule design.

Second change: the payoff-law correction, 9 September 2026. *Earlier claim:* the deployed game showed honest-plus-verified was the unique optimum, offered as evidence the rule works. *Source:* the notebook run during computational practice. *Revision:* the sweep returned $p^* = 0$, showing the game summed cost with the wrong

sign and penalised faking on benefit too, so honesty won regardless of any rule. The law was re-derived. *Consequence:* this is substantive, not cosmetic. The original game could not pose its own question, since faking was never tempting and there was nothing for a fine to deter. The corrected law gives a real threshold ($pF > 25$, so $p^* = 0.5$ at $F = 50$), which is the quantity the proposal needs. Affected: Section 3, the Section 5 table, the Open Science Statement, and rows 4–5 of Appendix E. A baseline that cannot fail its own test is not evidence.

Third change: honesty was modelled as a cost, not an asset, 9 September 2026. *Earlier claim:* the corrected law showed honesty could be an equilibrium, and I treated the analysis as complete. *Source:* reading the certification-design result of Mäkimattila et al. [6], in which a sender’s choice among offered tests is itself informative, together with Spence’s single-crossing condition [10]. *Revision:* in my law honest and faked traces were paid the same base benefit, so honesty won only by avoiding a penalty. I added a market premium Δ paid only for a verifiable trace file, making the condition $\Delta + pF > 25$. *Consequence:* the proposal now answers its own question. Before, it could show only that cheating does not pay; now it shows honesty can win because it is worth more, and that a premium of 25 removes the need to fine at all. Affected: Section 2 (hypothesis, model, Table 1), Q1, the Section 5 roadmap and the 2056 statement. *What remains unresolved:* Δ is still a parameter rather than a belief. If fakers can publish traces as cheaply as honest producers the signal pools and $\Delta \rightarrow 0$, so making Δ endogenous is the next study.

C Field and open-source inspiration

The *Joint Interdisciplinary Field Trip—Technology, Computation, Culture & Society in Action* took place on September 4, 2026.

Tencent Shanghai Office. Observation: WorkBuddy is Tencent’s AI agent platform, whose marketplace connects plugin developers with users. Interpretation: plugin quality and security are hard to observe before adoption, so buyers may rely on cheap signals, and costly verification can lead to a lemons market. Photograph: *WorkBuddy agent-plugin marketplace*, taken by the author, 4 September 2026.

Shanghai Science and Technology Museum. Observation: the 2023 Mathematics and Computer Science Prize panel [4] presents deep residual learning (He Kaiming et al.) through simplified diagrams. Interpretation: trust here rests on the museum’s institutional authority, not on independently checkable evidence. Photograph: *2023 Mathematics and Computer Science Prize panel*, taken by the author, 4 September 2026.

Two open-source tools. (i) *SLSA* [9], <https://slsa.dev/>, v1.2, accessed 6 September 2026; repository Apache-2.0. Capability: generates non-forgable build provenance—which commit built which artifact—so a verifier can check it. Limitation: it certifies the build chain, not whether the artifact does what its description claims. Consequence: SLSA supplies the build layer at near-zero cost, so my contribution is the claim layer it ignores. (ii) *CKAN* [3], <https://ckan.org/>, 2.11, accessed 6 September 2026; GNU AGPL-3.0. Capability: an open-data catalog with standardised metadata and public APIs. Limitation: it stores whatever publishers upload and never checks whether a dataset matches its description. Consequence:

CKAN is the natural substrate for the trace file, since a claim would deep-link to an exact dataset that the verifier recomputes.

D Review response and revision record

Status: pending. Peer reviews are scheduled for September 14, 2026 and have not been received; this section will be completed with the author response and the v2 revision. No reviewer feedback is claimed.

E Structured abstract and cumulative proposal map

Table 3: Appendix E structured abstract; the eight-row proposal structure.

Proposal element	Current statement
Background	Build-layer traceability already exists and generation is near-free, while unchecked quality uncertainty collapses markets. Foundations: Akerlof (1970); Buneman et al. (2001).
Motivation and application	Untested: which platform rules make honest trace files an equilibrium when they are free to generate and to fake. Scenario: a plugin marketplace setting the rules, with users and journalists as verifiers; it scales to public claims.
Three connected questions	Economics: which mechanisms—a fine pF or a premium Δ —make honesty a Nash equilibrium? Computation: what format and verifier make a per-claim declaration checkable at near-zero cost? Behavior: does traceability shrink the justification space, or does easy checking license less scrutiny? Integrated: which rules make honest disclosure stable when the evidence can be forged?
Method and model assumptions	Developers choose honest, none or fake, plus enrolment; the platform commits to (m, p, F) and the market sets Δ . Platform sets rules, developers act, spot-checks occur with probability p . Information is incomplete. Payoff is benefit plus Δ if honest, minus cost, plus stability if enrolled, minus pF if faking; a fake earns the same base benefit as an honest deliverable. Baseline nets 120; the computation is a 3×2 enumeration plus sweeps of p and Δ .
Results	Verified: with no rules and $\Delta = 0$ the fake wins (160 vs 135); a premium alone restores honesty ($\Delta = 26$), and without one honesty requires $pF > 25$ ($p^* = 0.5000$ at $F = 50$). The two substitute: $\Delta = 15$ gives $p^* = 0.20$; $\Delta = 25$ needs no enforcement. Five checks pass. Evidence type: simulated, from one shared payoff law. Two corrections were made; see Appendix B.
Intellectual merits	Checkability becomes an institutional variable rather than a technical property, extending lemons theory to claims under free generation, where the certification instrument is itself forgeable. The analysis separates enforcement from reward and shows they are substitutes, which reframes the design problem from policing to pricing.
Practical impacts	Private: lower screening costs on plugin and agent platforms. Public: auditable claims in museums, ministries and data portals. The SDG 4 tool lets students and the public learn why honest disclosure needs cheap verification, assessable by a short pre/post prediction question. The simulator is released; learning gains remain unevaluated.
Feedback received and revision made	September 7 workshop feedback established that build-layer provenance already exists, moving the contribution to the claim layer and revising the 2056 statement from cooperation under unobservability to a governed trust infrastructure. September 9, two dated changes: the sweep returned $p^* = 0$, exposing a sign error; and reading Mäkimattila et al. [6] showed honesty was modelled as a cost rather than an asset, so a premium Δ was added. Peer reviews pending; none claimed.